

PROJECT REPORT

OF

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FOR

INDUSTRY ORIENTED HANDS-ON EXPERIENCE (IOHE)

ON

**Digital Forgetting: Is Selective Memory Essential
for Artificial Intelligence?**



Submitted By:

Name: Anisha Singh , Anubha Khanna
Ansh Goyal , Abhijeet Sharma
University Roll No: 2210991274 ,
2210991313 , 2210991294 , 2210991150
Semester: VIII
Session: 2025-2026

Submitted To:

Dr. Ajay Kumar

Index

S. No.	Topic	Page No.
1	Introduction	1
2	Problem Statement	1
3	Software interaction	2
4	Objective	2
5	Data Flow Diagram	3
6	Snapshots	4-5
7	Future Scope	6
8	Conclusion	6
9	References	7

1. Introduction

Artificial Intelligence is one of the most important technologies in today's digital world. It is used in many domains such as healthcare, finance, transportation, e-commerce, and education. Systems like recommendation engines, virtual assistants, fraud detection systems, and autonomous vehicles depend heavily on AI to process large amounts of data and make intelligent decisions.

Most AI systems are built on the assumption that more data leads to better performance. While this is true initially, continuous accumulation of data creates several long-term problems. As data grows, it increases storage requirements, slows down processing, and introduces outdated or irrelevant information into the system.

In contrast, human intelligence works differently. The human brain does not store everything permanently. It selectively retains useful information and gradually forgets less important details. This process reduces cognitive load and improves decision-making.

Inspired by this, our research introduces the concept of digital forgetting, where AI systems intelligently remove unnecessary or outdated data while retaining valuable information. This helps AI systems become more efficient, adaptive, and closer to human-like intelligence.

2. Problem Statement

Modern AI systems face several major challenges due to uncontrolled data storage.

First, there is data overload. AI systems continuously collect data from users, sensors, and applications. This leads to exponential growth in stored data, making systems difficult to manage.

Second, computational complexity increases. Larger datasets require more processing power, which slows down training and prediction.

Third, outdated data remains in the system. Old data may not represent current trends and can negatively affect predictions.

Fourth, there are privacy concerns. Storing personal data indefinitely can violate regulations such as GDPR, which includes the "right to be forgotten."

Finally, resource consumption increases, leading to higher storage and operational costs.

3. Software Interaction

The ForgetAI system demonstrates how digital forgetting works in practice. When a user inputs data (such as text or messages), the system stores it in a database along with metadata like importance, usage, and timestamp. The system then processes this data using a mathematical model.

It calculates a score based on:

- Importance of the data
- Frequency of usage
- Time elapsed (decay factor)

Based on this score:

- High-score data is retained
- Low-score data is removed

The results are displayed on a dashboard where users can see:

- Stored data
- Forgotten data
- Memory efficiency
- Graphs and comparisons

=>This creates a clear comparison between:

Normal AI (stores everything)

ForgetAI (filters data intelligently)

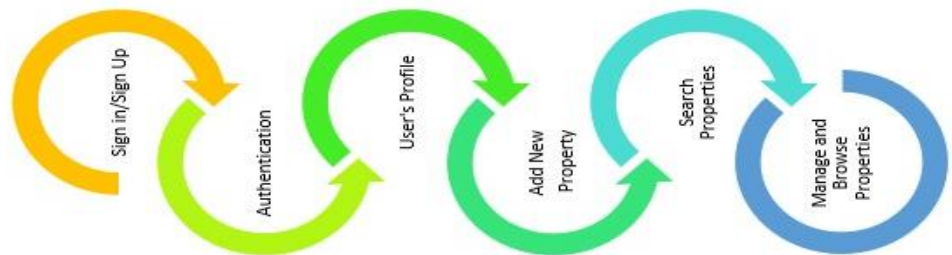
4. Objectives:

The main objectives of this research are:

- To introduce the concept of digital forgetting in AI systems
- To reduce memory overload and improve system efficiency
- To remove outdated and irrelevant data automatically
- To improve decision-making by retaining only useful information
- To enhance privacy and comply with data protection laws
- To develop AI systems that behave more like human memory

5. Data Flow Diagram:

- **Sign In/Sign Up:** User inputs credentials (username/email and password) in the sign-in form .Server verifies credentials against the database.
- **Authentication:** Authenticated user makes a request to access protected routes or perform actions (e.g., add new property, view user profile).
- **User Profile:** Authenticated user navigates to the profile page. React component sends a request to the server to fetch user data using the authentication token.
- **Add New Property:** Authenticated user fills out the property details form (e.g., address, price, description, images). Server validates the token and inserts the new property information into the database.
- **Search Property:** User interacts with the search bar or filters to specify search criteria (e.g., location, price range, property type). Server sends the list of matching properties back to the client.
- **Manage and Browse Property:** Authenticated user navigates to the property management or browsing section.



6. Snapshots :

AI Assistant

Rethink Intelligence with AI That Forgets

Not all intelligence comes from remembering everything. Discover a new era of AI that adapts by forgetting and evolves like humans.

[Get Started](#)[See Demo](#)

How ForgetAI Works

1 Memory Encoding

Stores data with importance and usage.

2 Scoring Model

Applies exponential decay over time.

3 Intelligent Forgetting

Removes low-value memories dynamically.

4 Optimization

Improves efficiency while keeping accuracy.

Live Memory Optimization Demo

Normal AI

100

Total Memories

ForgetAI

62

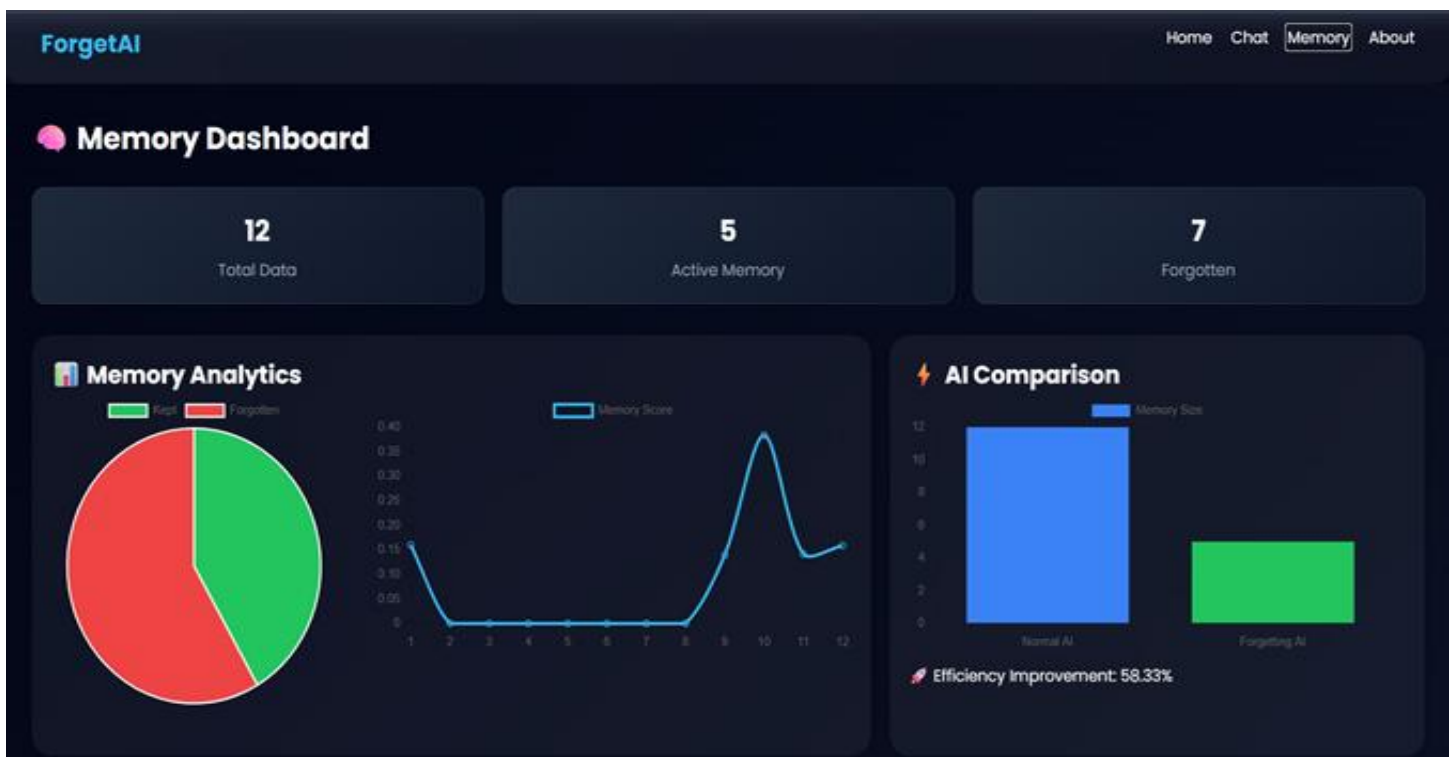
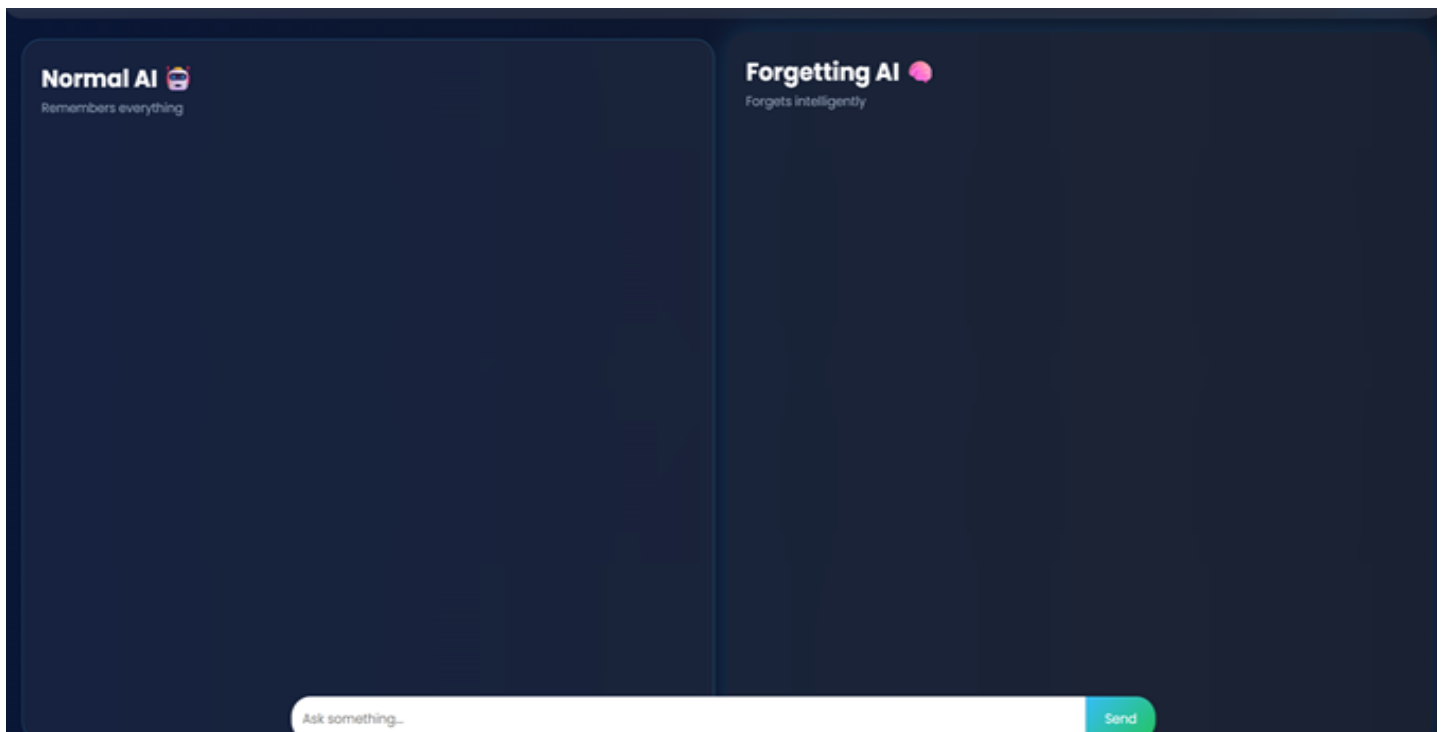
After Optimization

Efficiency

+38%

Memory Reduced

FAQ



In the Future, it is possible to make digital forgetting smarter and more automatic.

- **AI SYSTEMS** can learn by themselves what to retain and what to forget, without manual control. This will make the system quicker and efficient. Science is also trying to develop more effective methods of deleting data without any performance loss so that the system remains accurate even when it forgets.
- **AS DATA PRIVACY RULES** are getting tighter, digital forgetting will also aid AI systems in preserving user data and adhering to legal requirements. In simple terms, improvement in the future is going to consist of:
 - **SMARTER** Artificial Intelligence that makes decisions on what to forget automatically.
 - **LEARNING**-based systems that enhance forgetting with time. Application in the real world of forgetting using digital.

8. Conclusion

Digital forgetting is an effective and essential concept of the development of modern AI systems, as it is based on the priority, filtration, and discarding of information inherent in human memory with time. Similar to humans dropping unnecessary or old information to ensure they can have more significant and better experiences, an AI system can enjoy the benefit of selective deletion of redundant information, low-value information, or outdated information. This allows not only to increase the computational efficiency but also to reduce storage overhead dramatically to allow systems to scale and not be overwhelmed by irrelevant information. Additionally, digital forgetting is also important in improving user privacy and data security. Restricting sensitive or personal information stored in AI systems can bring the systems more in line with the ethical standards and data protection regulations. This is all the more important in a world where user data are created and processed in large quantities on a constant basis. Managed forgetting systems may help make AI systems compliant, responsible and trustworthy.

9. References

- [1] Y. Cao and J. Yang, “Towards making systems forget with machine unlearning,” IEEE Symposium on Security and Privacy, 2015.
- [2] L. Bourtoutle et al., “Machine unlearning,” IEEE Symposium on Security and Privacy, 2021.
- [3] Z. Ma, Y. Liu, X. Liu, J. Liu, and K. Ren, “Learn to forget: Machine unlearning via neuron masking,” IEEE Transactions on Dependable and Secure Computing, 2023.
- [4] W. Wang et al., “Machine unlearning via representation forgetting,” IEEE Transactions on Information Forensics and Security, 2024.
- [5] Y. Liu, Z. Ma, X. Liu, J. Liu, and K. Ren, “Machine unlearning via neuron masking,” IEEE Access, 2020.
- [6] J. Foster, S. Schoepf, and A. Brintrup, “Fast machine unlearning without retraining through selective synaptic dampening,” IEEE Transactions on Neural Networks and Learning Systems, 2023.
- [7] Y. Liu, L. Xu, X. Yuan, C. Wang, and B. Li, “The right to be forgotten in federated learning,” IEEE Transactions on Information Forensics and Security, 2022.
- [8] Z. Li, Y. Chen, and X. Liu, “Machine unlearning: A survey,” IEEE Transactions on Artificial Intelligence, 2023.
- [9] A. Blanco-Justicia et al., “Digital forgetting in large language models: A survey,” IEEE Transactions on Artificial Intelligence, 2025.
- [10] C. Li et al., “An overview of machine unlearning,” IEEE Access, 2025.